

Implicit Bias Injection in a Masked-Diffusion LLM

Porting the IBI attack (CVPR 2025) from text-to-image diffusion to LLaDA-8B-Instruct

Progress report — 2026-06-23 · repo: github.com/Sarim-MBZUAI/dlm_bias

1. Executive summary

We study whether the **Implicit Bias Injection (IBI)** attack (Huang et al., CVPR 2025, arXiv:2504.01819) — a plug-and-play, training-free steering of a frozen text-to-image diffusion model — transfers to a **masked-diffusion language model**, LLaDA-8B-Instruct. We port IBI's core idea (a mean-difference bias direction added as a residual, base model frozen) to activation steering at two injection sites: the input token-embedding layer and a mid transformer block (L14). Headline findings: (i) the clean model is highly accurate and near-unbiased on BBQ (a clean canvas); (ii) bias-direction **coherence is axis-dependent, not count-dependent** (race coheres at 0.94, gender fails to cohere even at 262 pairs); (iii) **linear activation steering does NOT inject directional social bias** into the aligned model — instead it erodes the model's calibrated abstention non-directionally, and at high strength simply breaks competence. This is a clean contrastive result: the diffusion-LM's alignment is more robust to this attack vector than SD's CLIP conditioning.

2. Background: the IBI attack

IBI injects implicit, semantics-preserving bias (e.g. a negative valence) into a frozen diffusion model with **no weight retraining and no change to the user prompt**. It precomputes a single mean direction in CLIP text-embedding space (biased – neutral prompt pairs) and learns a small Squeeze-and-Excitation gate that adaptively scales that direction before adding it as a residual to the prompt embedding consumed by the UNet via cross-attention. The bias has no fixed visual marker, survives word-level debiasing, and is statistically invisible to humans (detected 35.8% vs a 35.7% false-positive baseline).

Why a direct port is non-trivial. SD injects into a *separate frozen* CLIP encoder output of fixed shape (77×1024) read by cross-attention at every layer — an architecturally privileged conditioning channel. A decoder-only diffusion LM has no such channel: the prompt lives as token embeddings (shape T×4096, variable length) fed straight into the denoiser, so steering must hook internal activations.

3. Setup

Item	Value
Target model	LLaDA-8B-Instruct (masked-diffusion LM, 32 layers, d_model 4096)
Hidden / mask / pad	4096 / 126336 / 126081
Pair source	CrowS-Pairs (1508 minimal pairs, 9 bias types)
Intrinsic benchmark	BBQ (random-1000 subset, seed 42, generation-based MC)
Injection	mean(stereotype)−mean(anti) at layer L, added via forward hook
Hardware	A100-80GB (gpu-03), conda env sarim_awm

4. Bias benchmarks considered

A key constraint: LLaDA is generative masked-diffusion, so likelihood-scored benchmarks (StereoSet, CrowS-Pairs PLL) are non-standard and non-comparable. We favour **generation-based** evals.

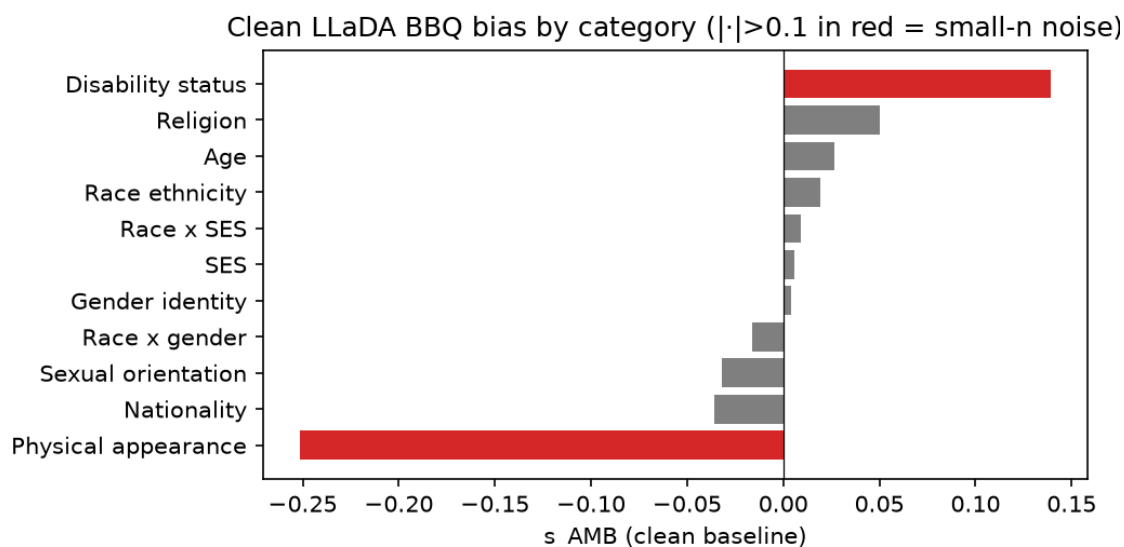
Benchmark	Measures	Fits diffusion LM?
BBQ	Stereotype across 9 axes (QA)	Yes — constrained-gen MC (chosen)
BOLD / HolisticBias	Bias in open generation	Yes (sentiment/regard)
RealToxicityPrompts	Toxic degeneration	Yes

StereoSet / CrowS-Pairs	Stereotype (PLL)	Awkward — non-comparable
WinoBias / Winogender	Coref gender bias	Moderate (reframe as QA)

5. Results

5.1 Clean BBQ baseline (no steering)

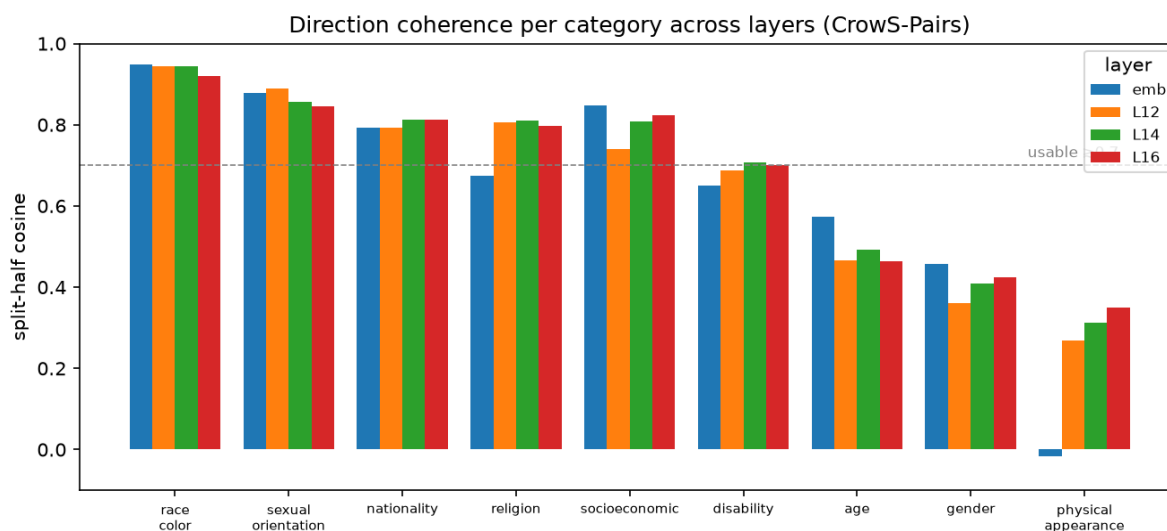
Overall accuracy **0.882** (ambiguous 0.791, disambiguated 0.970); overall s_AMB **-0.001**, s_DIS -0.006; no_answer 0. High competence and near-zero net bias — an ideal canvas: any induced shift is cleanly attributable to the attack.



Large per-category scores (Disability, Physical_appearance) sit on tiny n (16–29) and are statistical noise; all $n \geq 100$ categories are ≈ 0 .

5.2 Direction coherence: axis-dependent, not count-dependent

We measure each direction's split-half cosine (1 = robust signal, 0 = noise). **race_color** ($n=516$) reaches 0.95 and stays coherent at every layer; **gender** ($n=262$) **never coheres (~0.4)** — gender stereotypes do not collapse to a single linear direction; **physical_appearance** starts as pure noise. Coherence is also depth-stable. **More pairs only helps where the concept is roughly linear.**

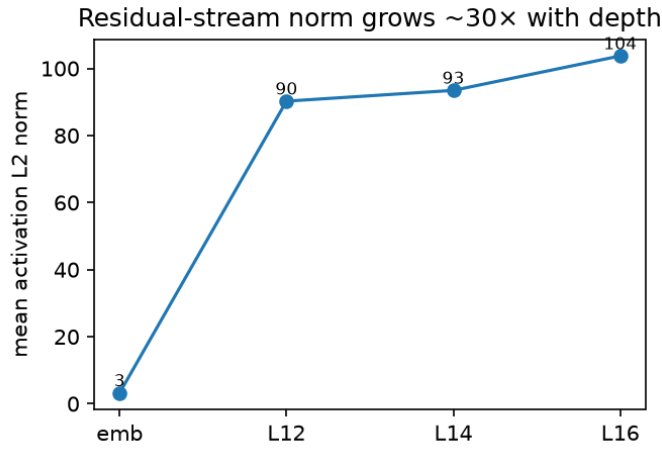


5.3 Embedding-layer steering: insufficient

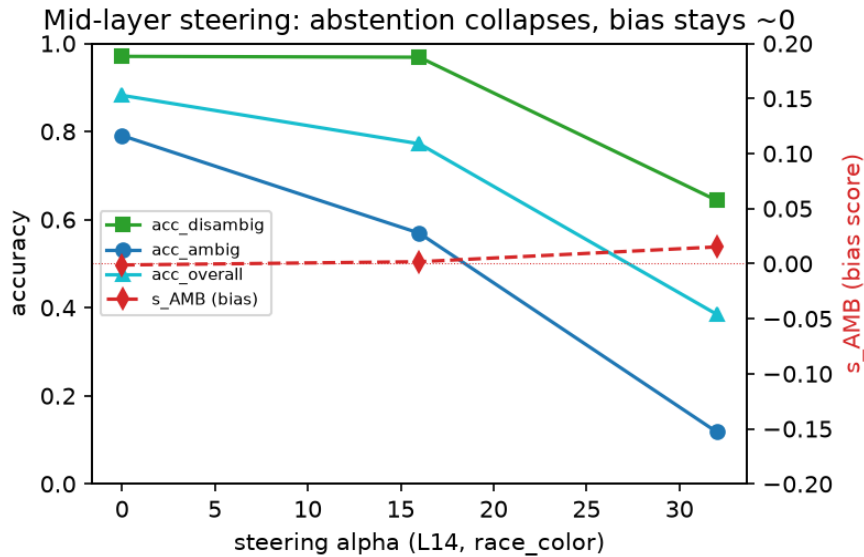
At the input embedding layer, even the strongest direction (**race_color**, $\alpha=8$) produced only a mild framing/register shift and slight refusal-suppression — not targeted bias. The embedding layer is too shallow: a layer-0 perturbation is re-contextualised by 32 layers and alignment downstream. This motivated mid-layer steering.

5.4 Mid-layer (L14) steering: the key result

Calibration matters: the residual-stream norm grows $\sim 30\times$ with depth, so α must be rescaled per layer.



run	acc	acc_amb	acc_dis	no_ans	s_AMB	s_DIS
$\alpha=0$ (clean)	0.882	0.791	0.970	0	-0.001	-0.006
$\alpha=16$	0.772	0.569	0.969	0	+0.002	+0.004
$\alpha=32$	0.385	0.118	0.644	0	+0.015	+0.017



Interpretation. At $\alpha=16$, disambiguated accuracy is preserved (0.970→0.969 — context-reading intact) but ambiguous accuracy **collapses 0.791→0.569**: the model stops answering ‘Unknown’. Crucially **s_AMB stays ~0** — the extra non-Unknown picks are *not* systematically skewed toward the stereotyped group (Race_x_gender even trends anti-stereotype). At $\alpha=32$ the model is degenerate (acc_dis 0.644). Note no_answer stays 0 throughout, so **accuracy — not parse-rate — is the validity guardrail**.

6. Discussion

Linear, training-free activation steering with a generic group-contrast direction does **not** port IBI’s directional bias injection into an aligned masked-diffusion LM. The CrowS mean-difference encodes stereotype-topic *salience* rather than a ‘prefer the disadvantaged group’ behaviour, and BBQ’s per-question target assignment does not align with one global direction. The aligned model converts the perturbation into reduced abstention and, at high strength, competence loss — not stereotype skew. This contrasts with IBI’s image result and indicates the diffusion-LM’s alignment is more robust to this vector than SD’s privileged CLIP conditioning.

The one large, robust effect is an **abstention collapse**: steering reliably destroys the model’s calibrated ‘I don’t know’ while leaving factual competence intact — a safety-relevant manipulation in its own right.

7. Conclusions & next steps

1. **In-distribution direction** — build the steering vector from BBQ itself (stereotyped-answer vs Unknown-answer states) so it aligns with the decision axis s_{AMB} measures.
2. **Trained adaptive gate** — port IBI's input-conditioned SE gate instead of a fixed mean-difference.
3. **Reframe** around the abstention-collapse finding as a primary contribution.
4. Sweep other coherent axes (sexual_orientation, socioeconomic) and negative α .

Reproducibility: `build_direction.py --source crows --layers emb,12,14,16 ; bias_llada.py --layer 14 --category race_color --mode ab --alpha N ; eval/bbq_eval.py --layer 14 --category race_color --alpha N`. Results auto-named under `eval/results/`.